$f(x) = \int f'(x)\,dx$

End-semester Examination Jan-May 2025
BT307 Biological Data Analysis

Time: 3 hr

Marks: 40

Instructions: 1) Preferably write the answers following the order of questions. 2) You MUST SHOW ALL the relevant steps of your calculation/derivation. 3) Clearly MARK the FINAL answer. 4) Use standard mathematical notations/symbols only. 5) Marks will be deducted for irrelevant calculations/derivations. 6) No marks for a partially correct answer.

Section A (each question has two marks)

Q1. X and Y are independent variables, and $\mathrm{var}(X) = \mathrm{var}(Y) = 1$. A and B are two data points in the X-Y plane. The coordinates of A and B are (2, 3) and (3, 2) respectively. Calculate the Mahalanobis distance between A and B.

$\sqrt{2}$

Q2. We performed PCA on 6-dimensional data. The eigenvalues of the covariance matrix are 5, 0.5, 1, 3, 1.5, 4. Calculate the fraction of the total variation in data explained/represented by the first two Principal Components.

0.333 0.267

Q3. The PMF of a discrete random variable X is $P(X=k) = \frac{4^k}{k!}e^{-4}$. Here X ≥ 0. Calculate the variance of X.

Q4. A logistic regression model is built to predict the likelihood of developing a certain disease based on three predictors: x, y, and z. The model equation is:

$$p(x,y,z) = \frac{1}{1+e^{-(-4+0.1x+0.1y+0.1z)}}$$

A patient has the following profile: x = y = z = 20. Calculate the odds that this person is at risk of developing the disease.

e^2

$\frac{153.5}{76.75}$

Q5. Calculate the slope of the line fitted through linear regression for the given data.

X	3	7	10	15
Y	8	16	22	32

$\bar{x} = 8.75$ XY: 66.125, 6.125, 3.125, 78.125

$\bar{y} = 19.5$ X^2: 33.0625, 3.0625, 1.5625, 39.0625

$x-\bar{x}$ −5.75 −1.75 1.25 6.25

$y-\bar{y}$ −11.5 −3.5 2.5 12.5

$(x-\bar{x})^2$

≈ 2.071.97

Section B (each question has six marks)

Q6. A random variable X follows uniform distribution X ~ U(p, q). Derive its CDF. You must show all the steps of the derivation and provide adequate explanations.

Var(1 − p)

Var p

10.868 − 8.892

−95.304 + 97.34

Q7. Let X and Y are two variables. The Pearson's correlation between X and Y is r_{XY}. We performed a linear regression, considering X and Y independent and dependent variables. Derive the relation between the coefficient of determination of this regression and r_{XY}. You must show all the steps of the derivation and provide adequate explanations.

$r^2 = R^2$

Q8. Consider a three-class classification problem (classes A, B, and C) solved via a one-vs-all logistic regression scheme. We created classifiers that use the following logistic equation:

$$\sigma(x) = \frac{1}{1+e^{-z(x)}}$$

0 1.5

Here, x is the predictor. For A vs Others: $z(x) = x - 1$. For B vs others: $z(x) = 2 - 0.5x$. For C vs others: $z(x) = 0.5 + 0.2x$. Determine the class where a new observation with $x = 1$ will be assigned.

0.7 B

Q9. The mean and standard deviation of the population's fasting blood glucose level are 100 mg/dL and 6 mg/dL, respectively. We randomly sampled nine persons and measured their fasting blood glucose levels. Let $\bar{G}$ be the mean fasting blood glucose levels of these people. What is the probability that $98 \le \bar{G} \le 102$?

0.68

Q10. We performed PCA for the data shown here. The loading matrix is given. Project the data on Principal Components 1 and 2 (PC1 and PC2). Calculate the Euclidian distance between samples 1 and 2 in the PC1-PC2 space.

Data:

	Drug 1	Drug 2	Drug 3
Sample 1	1	2	3
Sample 2	4	5	6
Sample 3	1	2	3
Sample 4	2	4	6

Loading matrix:

$$\begin{bmatrix} 1 & 1 & 3 \\ 2 & 1 & 5 \\ 3 & 1 & 8 \end{bmatrix}$$

$9\sqrt{5} = 20.12$

$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}$

1+4+9 1+2+3

4+10+18 4+5+6

14 6

15

32